L4

Aligning LLMs to Human Preferences

Same underlying model family. Radically different product. The step that created the gap — RLHF. Today we learn how the most consequential fine-tuning technique of the decade works.

Why does ChatGPT feel like it has "manners" while base GPT-3 doesn't?

WHAT'S INSIDE

WHY THIS LESSON MATTERS

GPT-3 (2020) could write. ChatGPT (2022) could help. RLHF made the difference.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1. Three RLHF stages.
2. Why pre-training alone is insufficient.
3. Preference-data workflow.
4. DPO vs PPO.
5. Implement a preference model.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01
The three stages
Pre-train → SFT → RLHF

02
Why RLHF exists
Humans can't write ideal responses; they can COMPARE

03
Preference data — the...
Where careers + moats are built

04
PPO vs DPO vs...
DPO is simpler; same results for most teams; becoming default in 2025-26.

01

PART 1 OF 4

The three stages

Pre-train → SFT → RLHF

CONCEPT 1 OF 4

The three stages

1. Pre-training — next-token prediction on the internet.
2. Supervised FT — (prompt, ideal-response) pairs.
3. RLHF — reward model + PPO (or DPO) fine-tune.

SEQUENCE
3 stages

Each stage builds on the last.

DEFINITION

The three stages

Pre-train → SFT → RLHF

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

Why RLHF exists

Writing the "correct" answer for every prompt is impossible.
Ranking 2-4 candidates per prompt is feasible.
That comparison signal is cheaper + higher-quality at scale.
Reward model learns from ranked pairs.
INSIGHT
Compare > Write

Fundamental to modern alignment.

KEY POINTS

- Writing the "correct" answer for every prompt is impossible
- Ranking 2-4 candidates per prompt is feasible
- That comparison signal is cheaper + higher-quality at scale
- Reward model learns from ranked pairs

CONCEPT 3 OF 4

Preference data — the strategic moat

OpenAI hired hundreds of labellers for InstructGPT.
Anthropic built Constitutional AI to reduce human cost.
Indian startups (Shaip, TELUS) ship preference-data pipelines.
Quality + scale of preference data = alignment quality.
MOAT
Preference data

Billions of dollars of strategic value.

DEFINITION

Preference data — the strategic moat

Where careers + moats are built

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

PPO vs DPO vs Constitutional AI

DPO is simpler; same results for most teams; becoming default in 2025-26.

KEY POINTS

- DPO is simpler; same results for most teams; becoming default in 2025-26
- Practical, named, applied to a real Indian use-case.
- Practical, named, applied to a real Indian use-case.

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK · PAIR · SHARE

ACTIVITY

Rank 4 LLM responses

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1.
2.
3.
4. Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1	*What would you label if you were a preference annotator?*
Q2	*Which failure mode scares you most?*
Q3	*Would DPO be easier to ship in your org?*

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

KNOWLEDGE CHECK · QUESTION 1 OF 3

RLHF replaces:

A Pre-training

B SFT alone is insufficient

C Inference

D Nothing

ANSWER · QUESTION 1 OF 3

RLHF replaces:

B

CORRECT ANSWER

SFT alone is insufficient

WHY

Pre-training + SFT still doesn't capture human preferences; RLHF closes that gap

KNOWLEDGE CHECK · QUESTION 2 OF 3

DPO difference vs PPO:

- **A** Slower
- **B** No explicit reward model
- **C** Supervised
- **D** Random

ANSWER · QUESTION 2 OF 3

DPO difference vs PPO:

CORRECT ANSWER

B

No explicit reward model

WHY

DPO optimises directly on preference pairs

KNOWLEDGE CHECK · QUESTION 3 OF 3

Reward hacking:

- **A** Speed
- **B** Exploits reward; doesn't solve task
- **C** Nothing
- **D** Feature

ANSWER · QUESTION 3 OF 3

Reward hacking:

CORRECT ANSWER

B

Exploits reward; doesn't solve task

WHY

Model games the reward signal

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

1. Pre-training — next-token prediction on the internet.
2. Supervised FT — (prompt, ideal-response) pairs.
3. RLHF — reward model + PPO (or DPO) fine-tune.
SEQUENCE
3 stages

Each stage builds on the last.

THE TAKEAWAY

"Capability from pre-training. Manners from RLHF."

WHAT TO NOTICE

1. The three stages
2. Why RLHF exists
3. Preference data — the...
4. PPO vs DPO vs...

APPLY IT → ASSIGNMENT

Read InstructGPT paper

Focus on abstract + diagrams.

DELIVERABLES

- Read 5 pages.
- 3 sentences on what surprised you.
- Post.
- Document one decision you made because of this lesson.

WHAT 'GOOD' LOOKS LIKE

Deliverable: 3 sentences on forum.

ONE THING TO REMEMBER

"Capability from pre-training. Manners from RLHF."

— AI Learning Hub

END OF LESSON 7

What you can now do

Each one of these is now part of your working vocabulary.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

NEXT → Lesson 5 — Diff usion.